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Safety Data Sheet

1. Chemical product and company identification

Product name : Titanium(Ⅲ) chloride solution

Company information

Name of manufacturer : KANTO CHEMICAL CO., INC.
Address : 2-1, Nihonbashi, Muromachi 2-Chome, Chuo-Ku, Tokyo, 103-0022, JP
Name of section : Reagent Division, Business Administration Department
Telephone number : +81-3-6214-1090
Facsimile number : +81-3-3241-1047
Mail address : BC32@gms.kanto.co.jp
Reference No : 40172

2. Hazards identification

GHS classification

Health hazards	Acute toxicity (inhalation:dust/mist)	Category 4
	Skin corrosion/irritation	Category 1C
	Serious eye damage/eye irritation	Category 1
	Respiratory sensitization	Category 1
	Specific target organ toxicity (single exposure)	Category 2 (respiratory organs)
	Specific target organ toxicity (repeated exposure)	Category 2 (teeth, respiratory organs)
Environmental hazards	Aquatic acute	Category 2

Hazard pictograms



Signal word : Danger

Hazard statements :
Causes severe skin burns and eye damage
Harmful if inhaled
May cause an allergy or asthma symptoms or breathing difficulties if inhaled
May cause damage to organs (respiratory organs)
May cause damage to organs (teeth, respiratory organs) through prolonged or repeated exposure
Toxic to aquatic life

Precautionary statements

Prevention :
Do not breathe mist/vapors.
Wash hands, forearms and face thoroughly after handling.
Do not eat, drink or smoke when using this product.
Use only outdoors or in a well-ventilated area.
Avoid release to the environment.
Wear protective gloves/protective clothing/eye protection/face

protection.
Wear respiratory protection.

Response : IF SWALLOWED: Rinse mouth. Do not induce vomiting.
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water .
IF INHALED: Remove person to fresh air and keep comfortable for breathing
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
IF exposed or concerned: Call a POISON CENTER or doctor.
Immediately call a POISON CENTER or doctor.
Call a POISON CENTER or doctor if you feel unwell.
Get medical advice/attention if you feel unwell.
If experiencing respiratory symptoms: Call a POISON CENTER or doctor.

Storage : Store locked up.

Disposal : Dispose of contents/container to hazardous or special waste collection point, in accordance with local, regional, national and/or international regulation.

3. Composition/information on ingredients

Distinction of substance or mixture : Mixture

Chemical name	Concentration (%)	Formula	TSCA	EC-No.	CAS RN
Titanium(III) chloride	≥ 20	TiCl3	Listed	231-728-9	7705-07-9
Hydrogen chloride	≈ 9	HCl	Listed	2315957	7647-01-0

4. First aid measures

First aid measures

First-aid measures after inhalation : Remove the victim to fresh air, and make him blow his nose and gargle. If necessary, get medical treatment.

First-aid measures after skin contact : Wash the affected areas under running water, get medical treatment as soon as possible.

First-aid measures after eye contact : Wash the affected areas under running water for at least 15 minutes. Get medical treatment.

First-aid measures after ingestion : Rinse mouth with water. Give the victim one or two glasses of water or milk. Do not induce vomiting. Get medical treatment as soon as possible.

Personal Protection in First Aid and Measures : Rescuers should wear proper protective equipment like rubber gloves, goggles.

5. Fire fighting measures

Suitable extinguishing media : This product is noncombustible.

Unsuitable extinguishing media : None

Fire hazard : When heated, corrosive hydrogen chloride gas is generated.

Firefighting instructions	:	Move containers from fire area if it can be done without risk, if not possible, apply water from a safe distance to cool and protect surrounding area.
Personal protection (Emergency response)	:	Firefighters should wear protective equipment.

6. Accidental release measures

Personal Precautions, Protective Equipment and Emergency Procedures

General measures	:	Wear proper protective equipment and avoid contact with skin and inhalation of vapor. Conduct operations from upwind and evacuate people downwind. Keep away personnel except for authorized ones from spillage area by stretching ropes.

Environmental precautions

Environmental precautions	:	Attention should be given to avoid discharge of spilled product into rivers and resulting environmental damage. When diluting spill with large amounts of water, discharge of untreated wastewater into the environment must be avoided.

Methods and Equipment for Containment and Cleaning up

For containment	:	Absorb spill with diatomaceous earth or dry sand and place in container. Neutralize residue with calcium hydroxide solution or sodium carbonate solution and then flush with copious amounts of water.

7. Handling and storage

Handling

Technical measures	:	Wear proper protective equipment to avoid contact with skin or inhalation of vapor.
Precautions for safe handling	:	Use with an enclosed system or a local exhaust ventilation. Use in well-ventilated areas.

Storage

Storage conditions	:	Store in a dark, cool place and tightly closed.
Material used in packaging/containers	:	Glass, Fluorine resin, Polyethylene. Do not use glass, fluorine resin, polyethylene mild steel, copper, alminum. etc.

8. Exposure controls / Personal protection equipment

Hydrogen chloride	
ACGIH Ceiling	2 ppm

Appropriate engineering controls	:	Use with an enclosed system or a local exhaust ventilation.

Protective equipment

Respiratory protection	:	If necessary, wear a chemical cartridge respirator with acidic gases.
Hand protection	:	Impervious protective gloves
Eye protection	:	Safety goggles
Skin and body protection	:	Protective clothing, protective boots

9. Physical and chemical properties

Physical state	:	Liquid
Color	:	Dark violet
Odor	:	Pungent
pH	:	Strong acidity
Melting point	:	No data available
Freezing point	:	No data available
Boiling point	:	No data available
Flash point	:	No data available
Auto-ignition temperature	:	No data available
Decomposition temperature	:	No data available
Flammability (solid, gas)	:	Non flammable
Vapor pressure	:	No data available
Relative density	:	No data available
Specific gravity / density	:	1.23 g/cm^3 (20℃)
Relative gas density	:	No data available
Solubility	:	Water: Miscible
Partition coefficient n-octanol/water (log Pow)	:	No data available
Explosive limits (vol %)	:	No data available
Viscosity, kinematic	:	No data available
Particle characteristics	:	No data available

10. Stability and reactivity

Reactivity	:	Has strong reducing property.
Chemical stability	:	Stable under normal conditions.
Possibility of hazardous reactions	:	It reacts explosively when it comes in contact with an oxidant. Corrodes many metals.
Conditions to avoid	:	Light, heat.
Incompatible materials	:	Alkaline substances, oxidizing substances, metals.
Hazardous decomposition products	:	Chlorine, hydrogen chloride, titanium oxides.

11. Toxicological information

Acute toxicity (oral)	:	No classification rat LD50=238-277mg/kg (as hydrogen chloride) ＊20% of the mixture consists of ingredient(s) of unknown toxicity.
Acute toxicity (dermal)	:	No classification rabbit LD50>5010mg/kg (as hydrogen chloride) ＊20% of the mixture consists of ingredient(s) of unknown toxicity.
Acute toxicity (inhalation)	:	No classification (gas) Classification not possible (vapor) Harmful if inhaled (dust, mist) rat LC50=0.42mg/L/4h (as hydrogen chloride) ＊20% of the mixture consists of ingredient(s) of unknown toxicity.

Skin corrosion/irritation	:	Causes severe skin burns In an acute transdermal toxicity test of rats using a 20% titanium (III) chloride hydrochloric acid solution (hydrochloric acid concentration 2-3%), extremely strong corrosive toxicity was observed, so the skin irritation / corrosiveness test was not performed. Based on the report, it was classified into category 1C.
Serious eye damage/irritation	:	Causes serious eye damage Since it is classified as a skin corrosive substance in the section of skin corrosive / irritating in this category, the evaluation for the eyes was classified into category 1.
Respiratory sensitization	:	May cause an allergy or asthma symptoms or breathing difficulties if inhaled There is no information on titanium(Ⅲ) chloride. Based on the information on hydrogen chloride, this product was classified into category 1. Hydrogen chloride : Since the substance is on the sensitizing chemical substance list by Japanese Society of Occupational and Environmental allergy, the substance was classified into category 1. There is a report that after exposure to a cleaning product that contained hydrogen chloride, the subject developed bronchospasm and still had marked asthma symptoms 1 year after exposure that were triggered by exercise and inhalation of trivial concentrations of irritants.
Skin sensitization	:	Classification not possible
Germ cell mutagenicity	:	Classification not possible In vivo test data is not available and cannot be classified. Regarding titanium (III) chloride, there are positive reports in the Ames test. Hydrogen chloride is negative in the Ames test and false positive due to low pH in the in vitro chromosomal aberration test.
Carcinogenicity	:	Classification not possible
Reproductive toxicity	:	Classification not possible A test in which one tenth of titanium(Ⅲ) chloride LD50 amount was orally administered to pregnant rats on days 1 to 3 of gestation showed significantly fewer subjects with embryos at 4 and 8 cell stages on day 5 of gestation compared to the control group. However, the classification was not possible because of insufficient data.
STOT-single exposure	:	May cause damage to organs (respiratory organs) There is no information on titanium(Ⅲ) chloride. Based on the information on hydrogen chloride, this product was classified into category 2 (respiratory organs). Hydrogen chloride : In humans, inhalation exposure caused symptoms such as dyspnea, laryngitis, bronchitis, bronchoconstriction, pneumonia and edema. Inflammation and necrosis of the upper respiratory tract and pulmonary edema have also been reported. In animal tests, toxic effects with morphological change of the lung and bronchus such as bronchitis with mucosal necrosis, pulmonary edema, hemorrhage, and thrombus were observed at dose levels within the guidance value range for category 1. Based on the information for humans and animals, the substance was classified into category 1 (respiratory organs).

STOT-repeated exposure	:	May cause damage to organs (teeth, respiratory organs) through prolonged or repeated exposure There is no information on titanium(Ⅲ) chloride. Based on the information on hydrogen chloride, this product was classified into category 2 (teeth, respiratory organs). Hydrogen chloride : There are 2 or more reports for humans in which repeated exposure caused tooth damage by erosion. Additionally, an increase in the frequency of chronic bronchitis was reported. Based on the information, the substance was classified into category 1 (teeth, respiratory organs).
Aspiration hazard	:	Classification not possible

12. Ecological information

Ecotoxicity

Aquatic acute	:	Toxic to aquatic life Daphnia magna EC50=0.492mg/L/48h (as hydrogen chloride) ＊20% of mixture consists of ingredient(s) of unknown hazards to the aquatic environment.
Aquatic chronic	:	No classification Hydrogen chloride is considered to be toxic because the aqueous solution becomes a strong acid, but in environmental water, the toxic effect is mitigated by the buffering action. ＊20% of mixture consists of ingredient(s) of unknown hazards to the aquatic environment.

Persistence and degradability

No additional information available

Bioaccumulative potential

No additional information available

Mobility in soil

No additional information available

Hazardous to the ozone layer

Ozone	:	Classification not possible

13. Disposal considerations

Ecology - waste materials	:	Precipitation method : Add the chemical in calcium hydroxide solution or sodium carbonate solution gradually to form precipitates. Filter the precipitates and bury in a landfill site approved for hazardous-waste disposal. Or entrust approved waste disposal companies with the disposal.
Contaminated container and packaging	:	In case of disposal of empty bottles, dispose bottles after removing the content thoroughly.

14. Transport information

International Regulations

Transport by sea(IMDG)

UN-No. (IMDG)	:	2869

Proper Shipping Name (IMDG)	:	TITANIUM TRICHLORIDE MIXTURE
Packing group (IMDG)	:	III
Transport hazard class(es) (IMDG)	:	8

Air transport(IATA)

UN-No. (IATA)	:	2869
Proper Shipping Name (IATA)	:	Titanium trichloride mixture
Packing group (IATA)	:	III
Transport hazard class(es) (IATA)	:	8
Marine pollutant	:	Not applicable

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Pollutant category	:	Z
MFAG-No	:	157

15. Regulatory information

Regulatory information with regard to this substance in your country or region should be examined by your own responsibility.

16. Other information

Data sources : Encyclopaedia Chimica, Kyoritsu Shuppan Co, Ltd. (1963) .
Handbook of Dangerous Substances Springer-Verlag Tokyo (1991) .
Handbook of 17221 Chemical Products, The Chemical Daily Co. (2021) .
NITE Chemical Risk Information Platform (NITE-CHRIP), National Institute of Technology and Evaluation.

The information contained herein is based on several references and the present state of our knowledge. However the SDS does not always cover all information about the product, handle the product carefully.The information is intended to ordinary usage, in case of particular handlings, conduct appropriate safety measurements.The information herein is only provision of information, and it does not represent a guarantee the properties of the product.The Safety Data Sheet(SDS) is prepared based on JIS Z7253.